

## Field with one Element

It is a classical result that any algebraic variety  $V$  can almost naturally be embedded into projective space, given a choice of where to put the projective coordinate. Given such a fixed choice (say 0 so that  $\mathbb{A}^n \cong U_0$ ), then there is a natural choice  $V \rightarrow \overline{V}$  as well with an inverse  $W \rightarrow W \cap U_0$ . If  $\mathbf{Var}_{\overline{K}} \rightarrow \mathbf{Sch}_{\overline{K}}$  the fully-faithful functor embedding varieties into schemes,  $(V, \mathcal{O}_V)$  maps to  $(\mathrm{Spec} \mathcal{O}_V, \mathcal{O}_V)$ , we get a similar completion for  $\overline{K}$ -schemes:

$$\mathrm{Spec} A \rightarrow \mathrm{Proj} A[t]$$

where the inverse map is a little more difficult to define as must now really make a choice **insure of this**.

So there is a natural question of whether this is possible for affine schemes more generally, or perhaps even schemes. Unfortunately, this falls apart very quickly. Consider  $\mathrm{Spec} \mathbb{Z}$ . We would want to consider  $\mathrm{Proj} \mathbb{Z}[t]$ , however notice that the affine component of this is  $\mathrm{Spec} \mathbb{Z}[t]$ , so already we are getting something different. If we wanted to just figure out how to projective  $\mathbb{Z}$ , we would need some grading structure on  $\mathbb{Z}$ , however there isn't one to find! Thus that mean that all hope is lost? This article will explore some of the ideas in trying to fix this, which in part shall introduce the fun concept of a field with one element.

Consider the fact that every prime number in  $\mathbb{Z}$  has a natural corresponding non-trivial valuation  $\nu_p$ . This valuation naturally allows us to zoom in at every point. Furthermore,  $k[t]$  has an analogous result for each  $t - a$ ; by taking the collection of all non-trivial valuations of  $k[t]$  we end up with  $t - a$  for all  $a \in k$  as well as  $1/t$ . Well,  $\mathbb{Z}$  also has an evaluation at infinity, namely the Archimedean valuation. Perhaps this will solve our

$F_{\mathrm{un}}$  where the subscript “un” comes from the french word for *one* which was chosen so that  $F_{\mathrm{un}}$  reads in English as “fun”.